NAME: ______________________ **PERIOD:** ________ **DATE:** ____________

Which Regression Line Fits?

AP Statistics · Unit 5 · Topic 5.5 · B: 2027-02-12 · E: 2027-04-09

[STAR] TODAY'S PROBLEM

Suppose a solar-farm analyst uses hours of direct sunshine, x, to predict daily generation, y, in megawatt-hours. Summary statistics for 12 observed days are shown.

Quinn: "Using $b = r(s_y/s_x)$ gives $\hat{y} = 5.6 + 3.2x$. It passes through $(7, 28)$, which by itself proves it is the LSRL. Since $r = 0.80$, the model explains 80 percent of the variation."

Rowan: "Using $b = r(s_x/s_y)$ gives $\hat{y} = 26.6 + 0.2x$. It also passes through $(7, 28)$, so that line is equally eligible. Explained variation is 64 percent, and a zero-hour intercept needs caution because 0 is outside the observed range."

Solar-farm summary (12 observed days)

x range	$\bar{x}$	$\bar{y}$	s_x	s_y	r
4–10 h	7 h	28 MWh	1.5 h	6 MWh	0.80

FIRST TAKE — before the video (5 min)

Which clauses in Quinn's and Rowan's analyses look defensible? Cite one formula or line property.

[BOOK] MAKE THE REGRESSION PROPERTIES AGREE (keep on the page)

The least-squares regression line (LSRL) minimizes the sum of squared residuals and contains $(\bar{x}, \bar{y})$. Its slope is $b = r(s_y/s_x)$, its intercept is $a = \bar{y} - b\bar{x}$, and r^2 is the proportion of response variation explained by the linear model. Interpret slope as a predicted response change per unit of x. The intercept predicts at $x = 0$, which may be outside the observed range and unsupported in context.

Finish after the video:

DOK 1 **(a)** State what the LSRL minimizes, the point it contains, and the contextual meanings of slope and intercept.

DOK 2 **(b)** Calculate the LSRL slope and intercept, r^2, and the prediction at 8 sunshine hours. Verify the mean-point property.

DOK 3 **(c)** Evaluate both analyses and write one warranted synthesis. Coordinate the slope ratio and units, least-squares and mean-point properties, coefficient meanings, and the observed domain. Compare both lines at 8 hours and quantify the consequences of using the rejected slope and using r instead of r^2.

EXIT — turn this in

One thing the video changed about my first take: ______________________________